

Sri Lanka Institute of Information Technology



2025-Y2-S1-MLB-WE1G2-08

Web Based Transportation System

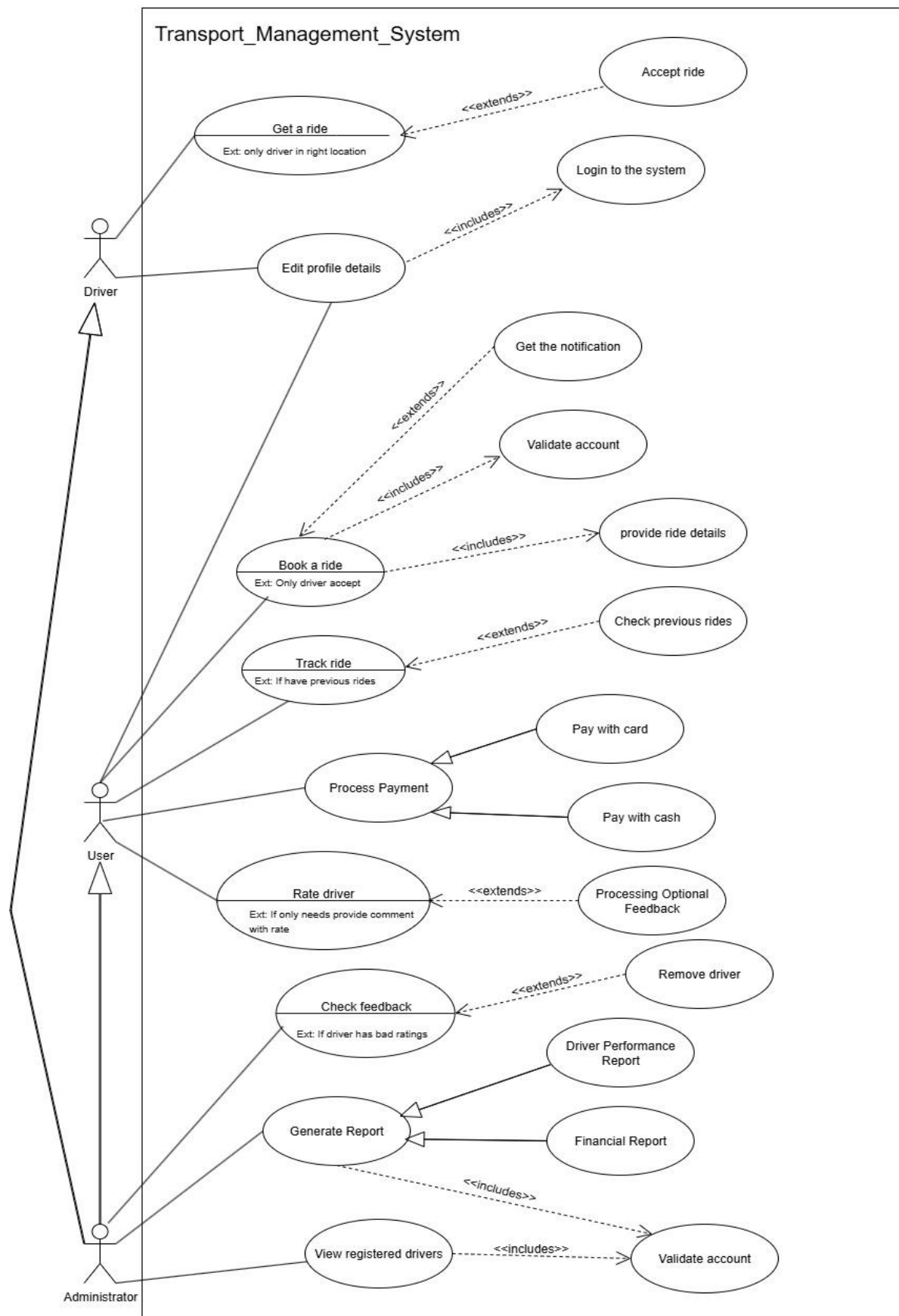
SE2030 - Software Engineering

B.Sc. (Hons) in Information Technology

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Use Case Diagram for Transport Management System, covering all major functions and users.



Note: We will assume that the admin can test both driver and user functions; therefore, the admin is the most prioritized person.

Link: <https://drive.google.com/file/d/1HYQCjpadzyrGYZOINk1hep0MJVKwJU3j/view?usp=sharing>

Individual Use Case Scenarios

User Registration

Number	1	
Name	Register User	
Summary	Users can create an account to access the application.	
Priority	High	
Preconditions	Users need to create account	
Postconditions	User accounts are created, and the user is logged in.	
Primary Actor	User	
Secondary Actor	System	
Trigger	User initiates the registration process.	
Main Scenario	Step	Action
	1	User opens the app and selects the "Register" option.
	2	User enters personal details (name, email, password).
	3	User agrees to the terms and conditions.
	4	User submits the registration form.
	5	System validates the input and creates a new account.
	6	System sends a confirmation email to the user.
	7	User receives confirmation and is logged into the application
Extensions	Step	Branching Action
	1a	If the app fails to load, display an error message and prompt for a refresh.
	1b	If the user is already logged in, redirect them to the home screen instead of showing the registration option.
	2a	If the user enters an invalid email format, prompt them to correct it.
	3a	If the user does not check the terms and conditions, prompt them to agree before proceeding.
	4a	If the user cancels the registration process, return them to the previous screen without submitting data.
	4b	If the submission fails due to a technical issue, notify the user and allow them to retry.
	6a	If the email confirmation fails to send, log the error and notify the user.
Open Issues	1	How to securely handle password storage and user data?
	2	What verification steps are needed (e.g. email verification)?

Ride Booking

Number	2	
Name	Book a Ride	
Summary	Users can book a ride through the web application.	
Priority	High	
Preconditions	Users must be logged into the application.	
Postconditions	Ride is successfully booked, and the user receives confirmation	
Primary Actor	User	
Secondary Actor	Driver	
Trigger	User initiates a ride booking request.	
Main Scenario	Step	Action
	1	User opens the app and selects the "Book a Ride" option.
	2	Users enter pickup and drop-off locations.
	3	User reviews the fare estimate and confirms the booking.
	4	System confirms the booking and notifies the user of the driver's details.
Extensions	Step	Branching Action
	1a	If the app fails to load, display an error message and prompt for a refresh.
	1b	If the user is not logged in, prompt them to log in before booking a ride.
	2a	If the user does not enter a pickup location, prompt them to provide one.
	2b	If the drop-off location is invalid, notify the user and request a valid address.
	3a	If the fare estimate cannot be calculated, notify the user and suggest trying again later.
	4a	If the booking confirmation fails, notify the user and allow them to retry the booking.
Open Issues	1	What happens if the user cancels the ride after booking?
	2	How to handle payment processing for different payment methods?

Track Ride

Number	3	
Name	Track Ride	
Summary	Users can track the status and location of their booked ride.	
Priority	High	
Preconditions	Users must have a ride booked.	
Postconditions	Users can view the status and location of the driver.	
Primary Actor	User	
Secondary Actor	Driver, System	
Trigger	User selects the option to track their ride.	
Main Scenario	Step	Action
	1	Users open the app and navigate to the “Track Ride” Section
	2	System retrieves the status of the booked ride.
	3	System displays the driver's current location on the map.
	4	Users can see the estimated time of arrival.
Extensions	Step	Branching Action
	1a	If the user is not logged in, prompt them to log in to access the feature.
	1b	If the app fails to load, display an error message and suggest a restart.
	2a	If the user has multiple rides, allow them to select which ride's status to view.
	3a	If the driver's location cannot be determined, inform the user and provide an estimated location
	3b	If the map fails to load, display a message and allow the user to refresh the map.
	4a	If the ride is delayed, update the ETA and notify the user.
Open Issues	1	How to ensure accurate real-time location tracking?
	2	What measures are in place for privacy and data protection during tracking?

Driver Assignment

Number	4	
Name	Request Ride	
Summary	Drivers can accept ride requests from users through the application.	
Priority	High	
Preconditions	Driver must be logged into the application and available to take rides.	
Postconditions	Driver receives ride details and starts the navigation to the pickup location	
Primary Actor	Driver	
Secondary Actor	User, System	
Trigger	User requests a ride, and the system matches the request with available drivers.	
Main Scenario	Step	Action
	1	User requests a ride through the app.
	2	System identifies available drivers and sends a ride request to them.
	3	Driver receives the ride request notification with ride details (pickup location, destination).
	4	Driver reviews the ride details and chooses to accept or reject the request.
	5	If accepted, the system notifies the user and provides driver details.
	6	Driver starts navigation to the pickup location.
Extensions	Step	Branching Action
	1a	If the user has not set a pickup location, prompt them to enter one.
	1b	If the web app fails to process the ride request, notify the user to try again later.
	2a	If no drivers are available, notify the user and suggest trying again later.
	2b	If multiple drivers are available, notify the user of the estimated wait time for each.
	3a	If the ride details are incomplete, notify the driver and provide them with the necessary information.
	4a	If the driver rejects the request, the system sends the request to another available driver.
Open Issues	6a	If the driver is delayed, notify the user with an updated arrival time.
	1	How to handle multiple ride requests when drivers are busy?
	2	What measures are in place to ensure driver's safety during pickups?

Process Payment

Number	5	
Name	Process Payment	
Summary	Users can make payments for their booked rides through the application.	
Priority	High	
Preconditions	User must have completed a ride or be in the process of booking one.	
Postconditions	Payment is processed, and the user receives a confirmation.	
Primary Actor	User	
Secondary Actor	Payment Gateway, System	
Trigger	User initiates the payment process after the ride or during booking.	
Main Scenario	Step	Action
	1	User selects the payment option after completing their ride or during booking.
	2	Users choose a payment method (credit card, debit card, digital wallet, etc.).
	3	User enters payment details (card number, expiration date, CVV).
	4	User confirms the payment amount and submits the payment.
	5	System processes the payment through the selected payment gateway.
	6	System notifies the user of a successful payment and provides a receipt.
Extensions	Step	Branching Action
	1a	If the user has no saved payment methods, prompt them to add one.
	1b	If the payment option is unavailable, notify the user and suggest alternatives.
	2a	If the selected payment method is declined, notify the user and prompt them to choose another method.
	3a	If the user enters invalid payment details, prompt them to correct the information.
	4a	If the user cancels the payment process, return them to the previous screen without processing.
	5a	If the payment gateway is unavailable, notify the user and suggest retrying later.
	5b	If the payment fails, log in to the attempt and notify the user of the failure.
Open Issues	1	How to ensure secure handling of payment information?
	2	What refund policies are in place for canceled rides?

Rate driver

Number	6	
Name	Rate Driver	
Summary	Users can rate drivers after a ride to provide feedback.	
Priority	Medium	
Preconditions	User must have completed a ride.	
Postconditions	User's rating and feedback are recorded in the system.	
Primary Actor	User	
Secondary Actor	Driver	
Trigger	User completes a ride and is prompted to rate the driver.	
Main Scenario	Step	Action
	1	Users are prompted to rate the driver after the ride.
	2	User selects a rating (1 to 5 stars).
	3	User has the option to provide additional comments.
	4	System records the rating and feedback.
Extensions	Step	Branching Action
	1a	If the user does not receive a prompt, allow them to access the rating option from their ride history.
	1b	If the ride is canceled, notify the user that they cannot rate the driver.
	2a	If the user attempts to submit a rating without selecting one, prompt them to choose a rating
	3a	If the user chooses to provide comments, ensure there is a character limit for the feedback.
	3b	If the user does not want to provide comments, allow them to skip this step easily.
	4a	If the feedback is successfully recorded, provide a confirmation message to the user.
Open Issues	1	How to handle multiple ratings from the same user for different rides?
	2	How to encourage more users to participate in the rating process?
	3	How to handle cases where a user does not receive a prompt to rate the driver?

Generate Report

Number	7	
Name	Admin Generate Report	
Summary	Admins can generate reports on rides, users, and drivers for analysis and insights.	
Priority	Medium	
Preconditions	Admin must be logged into the admin panel.	
Postconditions	Report is generated and made available for viewing.	
Primary Actor	Admin	
Secondary Actor	System	
Trigger	Admin navigates to the reporting section and requests a report.	
Main Scenario	Step	Action
	1	Admin opens the admin panel and selects the "Generate Report" option
	2	Admin chooses the type of report
	3	Admin specifies the date range and any filters
	4	Admin submits the report request.
	5	System processes the request and generates the report.
Extensions	Step	Branching Action
	1a	If the admin panel fails to load, display an error message and prompt for a refresh.
	1b	If the admin lacks permission to generate reports, notify them of the restriction.
	2a	If no report types are available, display a message indicating this and suggest checking settings.
	2b	If the selected report type is invalid, prompt the admin to select a valid option.
	3a	If the filters specified are invalid, notify the admin and allow them to adjust the filters
	4a	If the admin cancels the report request, return them to the previous screen without processing.
	5a	If report generation fails, log the error and notify the admin to try again later.
Open Issues	1	How to ensure data accuracy and integrity in reports?
	2	What permissions are required for different admin roles when generating reports?

Edit Profile Details

Number	8	
Name	Edit Profile Details	
Summary	Users can update their personal information in their profiles.	
Priority	High	
Preconditions	Users must be logged into the application.	
Postconditions	User's profile details are updated and saved.	
Primary Actor	User	
Secondary Actor	System	
Trigger	User navigates the profile settings.	
Main Scenario	Step	Action
	1	User opens the web app and selects the "Profile" option.
	2	User chooses the "Edit Profile" option.
	3	User updates personal information (name, email, phone number etc.).
	4	User submits the changes.
	5	System validates the input and updates the profile.
	6	System confirms the update and displays the new profile details.
Extensions	Step	Branching Action
	1a	If the web app fails to load, display an error message and prompt for a refresh.
	1b	If the user is not logged in, redirect them to the login page.
	2a	If the user cancels the edit action, return them to the profile overview without changes.
	3a	If required fields are left blank, notify the user to complete them before submission.
	4a	If the user cancels the submission, return them to the profile overview without changes.
	4b	If submission fails due to a technical issue, notify the user and allow them to retry.
	5a	If input validation fails, display specific error messages for each invalid field.
Open Issues	1	How to ensure data privacy and security when updating profile details?
	2	What measures are in place for verifying changes made to sensitive information?

Individual Use Case Scenarios (mentioned member name and registration number)

Student ID	Name	Individual Use case Scenario
IT24101142	Sandamuthu A.A.N	Register User
IT24510021	Kulatunga W.M.H.P	Ride Booking
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IT24101209	Madusanka W. A. L. P	Rate driver
IT24100687	Sansiri P.S.T	Driver Assignment
IT24100892	Gamage G.A.G	Payment